



Srinivasa Ramanujan Institute of Technology (AUTONOMOUS)

Rotarypuram Village, B K Samudram Mandal, Ananthapuramu - 515 701

APPLICATION FOR STUDENT INNOVATIVE PROJECT UNDER SEED MONEY GRANT (SMG) SCHEME

1. Project title (in brief): **GREEN QR: SMART TREE INFORMATION SYSTEM**
2. Engineering Department Performing the Project: **COMPUTER SCIENCE AND ENGINEERING**
3. Class and Section: IV B. Tech & Section-A
4. Name of guide: **Dr. P. Veera Prakash** ([mobile number: 9885919829](tel:9885919829), [Email Id: veera.cse@srit.ac.in](mailto:veera.cse@srit.ac.in))
5. Name of Team Members:
 - Ms. A Chaithanya, RollNo.224G1A0513, Mobile No: 9391100594, Email id: 224G1A0513@srit.ac.in
 - Mr. B Shantha Vardhan, Roll No. 224G1A0597, Mobile No: 9063947845, Email id: 224G1A0597@srit.ac.in
 - Mr. V Likith Kumar, Roll No. 224G1A0542, Mobile No: 9121904800, Email id: 224G1A0542@srit.ac.in
6. Team Leader of the Project: Ms. A Chaithanya, Email id: 224G1A0513@srit.ac.in Mobile No: 9391100594

(Please specify name with emailed and mobile number of the Team Leader)

7. Date of commencement of the Project: **15-12-2025**
8. Probable date of completion of the project: **05-04-2026**
9. Has a similar project been carried out at your college or in any other college/institution? If "Yes", please give details. As to how you are attempting to improvise the technology.

Yes

Similar QR Code-based Tree Identification systems have been implemented in some colleges where scanning a QR code displays basic tree details. However, those systems were limited to static information display and did not include structured user interaction or management features. Our project improves the existing system by introducing secure authentication and a structured dashboard designed specifically for institutional use. It enhances usability, transparency, and data management within the campus environment.

Key improvements include:

- Secure Admin and @srit.ac.in restricted User login system
- Contribution reporting with admin approval and notification system

These enhancements make the system more organized, secure, and suitable for smart campus implementation.

10. If this project is an improvement/extension of an earlier project, please give the title of earlier project and they are in which it was carried out. Specify the improvements/modifications clearly.

“QR Code Based Tree Identification System” (2023–2024)

The earlier project mainly focused on generating QR codes and displaying tree information after scanning. It did not include structured login, dashboard features, or contribution tracking. Our project extends it by adding controlled authentication, dashboard-based navigation, and structured data management features.

Major modifications include:

- Addition of secure login and user dashboard
- Inclusion of admin approval workflow and reference link storage

These improvements convert a basic QR display system into a comprehensive Smart Tree Information Management System for campus use.

11. Is similar device/ machine(outcome of the project)commercially available? If "Yes", what is the cost per unit?

Please give details.

Yes

- **Cost Per Unit:** ₹5,000 – ₹15,000 (depending on hardware materials like weatherproof QR plates).
- **Details:** While generic QR code scanners exist, commercial "Smart Botanical Systems" often involve specialized weatherproof tags and subscription-based cloud databases. Your project is a localized, low-cost alternative that integrates specific institutional authentication (@srit.ac.in) and community-driven reporting (Newly Planted/Dead tree reports) which is not standard in generic commercial apps.

12. Objective of the project highlighting its importance (not more than half a page):

The objective is to digitize and manage botanical information within a campus environment through a secure, QR-based IoT ecosystem.

- To implement a secure login system for SRIT members using **Single Sign-On (SSO)** to access exclusive tree data.
- To provide an interactive dashboard that categorizes flora into **Medicinal, Flowering, Fruit, and Rare trees**, enhancing educational engagement.
- To create a community-driven monitoring system where users can contribute by reporting newly planted or dead trees, helping the administration maintain a real-time digital inventory.
- To eliminate the need for physical information boards, providing a sustainable and easily updatable digital "Tree Bio" including resources and reference links.

13. **Methodology:**

Principle of Functioning:

The system operates on a cloud-based mobile architecture. Each tree is assigned a unique QR code. When a user scans the code via the "Green QR" app, the application fetches specific data from the database. The system uses Google Single Sign-On for authentication, ensuring only authorized @srit.ac.in users can view detailed profiles and contribute reports. The dashboard uses filtering logic to sort tree types and keeps a "My Scanned Trees" history for user reference.

Hardware & Software:

The system utilizes durable, weatherproof QR-coded plaques installed on trees. The software backend manages two primary modules: an Admin Module for approving reports and managing tree data, and a User Module for scanning and searching. The data module is specifically designed to store tree names, scientific names, and external reference links while excluding dynamic data like "age" to ensure long-term accuracy.

Power/Maintenance Requirements:

As a software-heavy solution, it requires minimal physical power (only the user's smartphone and server hosting). Maintenance involves periodic updates to the cloud database and physical inspection of QR tags to ensure they remain scannable.

14. **Expected Outcome of the project:**

The successful deployment of a fully functional **Green QR** mobile application available for the SRIT campus community. A centralized, verified digital database of all campus trees, searchable by name and scientific classification. Increased student awareness regarding biodiversity through the "Tree Details" page which includes external resources for further reading. A streamlined reporting mechanism that allows the administration to act quickly on "Dead Tree Reports" and acknowledge "Newly Planted Tree Reports" through the notification system.

15. **Application field of the project:**

- **Educational Institutions:** Digitalization of campus flora for botany and environmental science.
- **Botanical Gardens & Parks:** Providing visitors with deep-dive information without physical clutter.
- **Smart Cities:** Managing public green spaces and encouraging citizen participation in reporting tree health.

- **Eco-Tourism:** Enhancing visitor experience in nature reserves.

16. Does the project proposed is relevant to any of the Industry or Institution in and around your area Yes/NO
Yes

17. If Yes, Please provide details of the Industry/institution and contact details :

The **Green QR** system is highly relevant to educational institutions and environmental organizations in Andhra Pradesh that focus on sustainable campus management and digital literacy.

- **Srinivasa Ramanujan Institute of Technology (SRIT):**
 - **Contact:** +91 88866 30040
 - **Email:** principal@srit.ac.in
 - **Relevance:** As the primary host institution, the project digitizes the campus flora, providing a live database for the Department of Humanities and Sciences.
- **Andhra Pradesh Forest Department (Anantapur Division):**
 - **Contact:** +91 85542 73064 (DFO Anantapur)
 - **Relevance:** The technology can be adopted for "Nagaravanams" (City Forests) to educate the public about biodiversity.
- **Green-Campus Initiatives (National Level):** Relevant to institutions seeking NAAC/NIRF accreditation under "Institutional Values and Best Practices."

(**Note:** Preference will be given to those projects relevant to the industry/institution. Hence be specific in giving detailed information). Is the industry extending support-technology/funds/use the final product, please specify.

18. In case of fabrication of related project, an engineering drawing with dimensions / detailed design Calculations(approximately) of the machine/device should be attached to the proposal.

For the Green QR system, the "fabrication" involves the UI/UX architecture and the physical placement of QR modules. The following design components are prepared: System Architecture Diagram: A detailed flowchart showing the interaction between the Mobile App, Google SSO Authentication, and the Cloud Database. User Interface (UI) Wireframes: Scaled layouts of the User Dashboard, the Tree Detail page (showing advantages/disadvantages), and the Admin Approval module. QR Plaque Design: Dimensions for the physical tags (e.g., $10\text{cm} \times 10\text{cm}$) made of weather-resistant materials to be mounted on tree trunks. Database Schema: Detailed table structures for tree metadata, user scan history, and contribution logs.

19. Can the idea of your project be patentable:

No

If No, provide details:

The Green QR project is designed as an open-access educational tool rather than a proprietary invention. Reasoning The project utilizes established technologies such as QR code generation, Single Sign-On (SSO) APIs, and cloud-based databases. While the implementation and the integration for the SRIT campus are unique and highly useful, the underlying components are considered standard software development practices.

Focus: Instead of a patent, the project aims for a **Copyright** on the specific source code and database structure, and the

publication of a research paper in a reputed journal (Scopus/SCI) to share the methodology with other educational institutions.

20. Budget (**break-up details should be given**) Enclose in a Separate Page

21. Any other details (Please specify):

(Name & Signature of Project Guide/Mentor)

(Signature of Head of the Department)

<i>Comments made by Research and Development Cell</i>	<i>Recommended the advance amount of</i> <i>Rs</i> <i>Signature of R&D Convener.</i>
<i>Sanctioning Authority</i>	<i>Principal/Head of the Institution</i>

Note:

- Coordinator should submit Two copies of the application and budget proposal to the Department. One copy shall be retained by the Department and other one copy shall reach the R&D Cell by the Department.
- R&D Cell shall examine the application thoroughly for recommending the advance amount.
- The project mentor and student batch should submit a report over the status of the project once for every month.
- At the end of the project, three copies of fund Utilization details and detailed report shall be submitted to the Department by the coordinator. One copy shall be retained by the Department and other two copies shall reach the R&D Cell by the Department.
- For the closure of the project Scopus/SCI/ESCI publication is mandatory.

Budget Estimation:

S. No.	Item Head	Estimated Cost (Rs.)	Fund to be arranged by students / other sources (Rs.)	Fund required from the Institute in INR Total (Rs.)
A	Non-recurring (Capital Items)			
1.	Weatherproof QR Plaques (Anodized Aluminum/Acrylic)	3000	0	3000
2.	Cloud Hosting & Database	2500	0	2500
3.	Domain Name Registration	800	0	800
4.	High-Resolution Camera for Data Collection	Exiting	0	0
5.	Mounting Materials	500	0	500
	Subtotal (A)	6,800	0	6,800

B	Recurring Items (General)			
1.	Manpower	4500	0	4500
2.	Consumables	500	0	500
3.	Contingencies	500	0	500
4.	Travel	300	0	300
5.	Overhead	480	0	480
	Subtotal (Recurring Items)	6280	0	6280
	Total amount (A+B)	13,080	0	13,080

Activity Plan:

S. No.	<i>Activity/ Milestone</i>	No. of Months
<i>1</i>	<i>Literature Survey</i>	2 Weeks
2	Data Collection	1 Week
<i>3</i>	<i>UI/UX Design</i>	4 Weeks
<i>4</i>	Coding & Backend	3 Weeks
5	Testing	2 Weeks
<i>6</i>	<i>Preparation of Thesis</i>	1 Week
<i>7</i>	<i>Paper Publication</i>	4 Weeks

